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Graded Assignment • 15 min

Due Aug 20, 11:59 PM IST

Your grade: 96%

Your latest: **96%** • Your highest: **96%** • To pass you need at least 50%. We keep your highest score.

Next item →

1. Which of the following are true about fully connected and convolutional neural networks? 1 / 1 point

- [ ] Convolutional layers are primarily used for non-image data.
- [x] Convolutional layers use weight sharing.

Correct
Correct. Convolutional layers share weights which helps in reducing the number of parameters and capturing spatial features.

- [ ] Both fully connected and convolutional layers can handle varying input sizes natively.
- [ ] Fully connected networks are more computationally efficient than convolutional networks for image classification.
- [x] Fully connected layers have a fixed-size input.

Correct
Correct. Fully connected layers require a fixed-size input which can be a challenge when dealing with images of varying sizes.

2. Which steps are necessary for preparing the MNIST dataset for training a GAN? 1 / 1 point

- [ ] Converting the images to RGB format
- [x] Splitting the dataset into training and testing sets

Correct
Correct! Splitting the dataset helps in evaluating the model's performance on unseen data.

- [x] Loading the dataset

Correct
Correct! Loading the dataset is a fundamental step before any processing or training.

- [x] Normalizing the image pixel values

Correct
Correct! Normalizing pixel values ensures that the data is in a suitable range for the model to process.

- [ ] Applying random rotations to the images

3. What is the benefit of using the MNIST dataset for training a convolutional GAN? 1 / 1 point

- ( ) It contains colored images, which are ideal for testing
- (•) It is a well-studied dataset with a simple structure
- ( ) The images are high-resolution, which enhances model accuracy
- ( ) It has a large number of classes, making it suitable for complex models

Correct
Correct! The MNIST dataset's simplicity and extensive use in research make it ideal for training and experimenting with GANs.

4. What is the primary purpose of the DCGAN generator function? 1 / 1 point

- ( ) To reduce the dimensionality of image data
- (•) To convert random noise into meaningful images
- ( ) To classify images into different categories
- ( ) To encode images into a latent space

Correct
Correct! The generator function transforms random noise into coherent images.

5. Identify the correct steps involved in modifying and testing the DCGAN generator model. 0.8 / 1 point

- [x] Using a different activation function for the final layer

Correct
Correct! The choice of activation function can influence the output of the generator model.

- [x] Changing the dimensionality of the noise vector

Correct
Correct! Altering the noise vector's dimensionality can affect the generated images.

- [ ] Adjusting the learning rate of the optimizer
- [ ] Implementing dropout layers in the generator
- [ ] Increasing the number of discriminator layers

You didn't select all the correct answers